

A Quantitative Analysis of Pricing Behavior in California's Wholesale Electricity Market During Summer 2000[†]

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During the Summer of 2000, wholesale electricity prices in California were nearly 500% higher than they were during the same months in 1998 or 1999. This price explosion was unexpected and has called into question whether electricity restructuring will bring the benefits of competition promised to consumers. The purpose of this paper is to examine the factors that explain this increase in wholesale electricity prices. We simulate competitive benchmark prices for Summer of 2000 taking account of all relevant supply and demand factors—gas prices, demand, imports from other states, and emission permit prices. We then compare the simulated competitive benchmark prices with the actual prices observed. We find that there is a large gap between our benchmark competitive prices and observed prices, suggesting that the prices observed during Summer 2000 reflect, in part, the exercise of market power by suppliers. We then proceed to examine supplier behavior during high-price hours. We find evidence that suppliers withheld supply from the market that would have been profitable for price-taking firms to sell at the market price.

INTRODUCTION

During the Summer of 2000, wholesale electricity prices in California were nearly 500% higher than they were during the same months in 1998 or 1999. This explosion of prices was unexpected (CEC, 2000) and has called into

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† This paper integrates and updates analyses contained in Joskow and Kahn (2001a and b).

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question whether electricity restructuring will bring the benefits of competition promised to consumers. Federal and State government officials have initiated investigations and issued reports about the behavior and performance of California's wholesale electricity market.¹ Unlike previous price spikes observed in other US wholesale electricity markets, the California experience has not been a transient phenomenon of a few days' duration, but a persistent series of events lasting from June 2000 through roughly mid-June 2001.²

The purpose of this paper is to examine the factors that explain this increase in wholesale electricity prices. It covers the Summer months of 2000, with more intensive analysis of the month of June 2000. June is a good month to scrutinize because most generating units should have recently returned from service from their Spring maintenance outages and future developments in supply and demand conditions that may have affected the valuation of hydroelectric resources and tradeable emissions permits are unlikely to have been anticipated by suppliers.³ There were a number of changes in supply and demand conditions in 2000 that would suggest that prices should have been expected to increase from the previous years: natural gas prices increased, demand increased, and power imports available to California decreased in 2000 compared to 1998 and 1999. The first objective of this paper is to determine how much of the observed price increases can be explained by these three "market fundamentals," *assuming* that the wholesale power market was perfectly competitive. We do so by simulating competitive benchmark prices given these supply and demand factors prevailing over the Summer of 2000 and then compare the simulated competitive benchmark prices with the actual prices observed. We find that while these three supply and demand factors can explain a portion of the observed increase in prices, there is still a large gap between the observed prices and simulated competitive benchmark prices.

1. Reports include FERC Staff Report (2000), Kahn and Lynch (2000), California Independent System Operator Department of Market Analysis (2000), California Power Exchange Corporation Compliance Unit (2000) among others.

2. FERC (1998) gives a detailed account of price spikes in Midwestern markets in 1998. Price spikes in the Eastern US during 1999 were related to reliability problems of various kinds (DOE, 2000).

3. Prices in California remained remarkably high in October and November, then reached unprecedented levels during December 2000 and remained at those levels through the Winter and Spring months of 2001. The latter part of this period was also accompanied by an order of magnitude increase in gas prices, the evaporation of imports from the Northwest, a large fraction of California's generating capacity was unavailable to supply due to planned or forced outages, some of which were mandated by environmental regulators, new regulatory interventions, and utility credit problems that may have made some suppliers reluctant to supply voluntarily. It is clear that by late 2000, the normal functioning of the wholesale electricity markets had completely broken down. Joskow (2001) discusses price movements and various government initiatives for this entire period.

The second objective of this paper is to determine whether and how much of the difference between benchmark and actual prices can be explained by the prices of tradeable permits for NO_x emissions. These emissions permits must be held by generating plants and other affected sources in the South Coast Air Quality Management District (SCAQMD) pursuant to the Regional Clean Air Initiatives Market (RECLAIM) program.⁴ The prices for these emissions permits increased dramatically during the Summer of 2000 compared to earlier periods. Including the emissions permit prices in the supply costs of those generators subject to RECLAIM increases competitive benchmark prices for electricity significantly, especially by the end of the Summer 2000. However, even after taking account of NO_x permit costs, during most of the Summer there remains a large gap between the simulated benchmark prices and actual market prices. We tentatively attribute this gap to market power and related market imperfections associated with the structure of California's wholesale electricity markets.

The final objective of this paper is to examine whether our attribution of the observed gap between benchmark competitive prices and actual prices is consistent with available data on supplier behavior. Even in a perfectly competitive market, prices may rise above the short-run marginal operating cost of the last unit to clear the market when demand must be rationed by prices above marginal cost to balance supply and demand in the face of capacity constraints. Therefore the diagnosis of market power should include both an analysis of price/marginal cost margins and a companion analysis of supplier behavior. Accordingly, we examine whether potentially profitable generating capacity was withheld from the market during high-price hours.

Our examination of the publicly available data shows that a significant amount of generating capacity produced much less energy than could have been produced at marginal costs below observed market clearing prices, after accounting for the maximum amount of generating capacity that may have been held in reserve by the California Independent System Operator (CAISO). Therefore, either the units were suffering from unusual operational problems or

4. We have not examined air quality regulations that may restrict production through command-and-control regulations. It is our impression that such regulations are not binding in California.

they were being withheld from the market to increase prices.⁵ Interestingly, the one supplier for which we do not find any significant evidence of withholding had apparently contracted most of the output of its capacity forward and would not have benefited by driving up spot market prices by withholding output.

A number of previous studies have examined wholesale electricity prices in California and found some evidence of market power, especially during high demand periods. Most of this analysis relies on confidential data available only to the CAISO or to the California Power Exchange (PX). In addition to extending this kind of analysis to the Summer of 2000, our paper provides three innovations. First, it relies on data which are generally available to the public rather than on confidential data available only to the CAISO or PX and their respective market surveillance committees. Accordingly, unlike earlier studies, our results can be reproduced by others. Second, previous analyses of wholesale market prices in California have not systematically taken into account the prices for NO_x emissions permits which generating plants located in the Los Angeles area must hold to cover their emissions of NO_x, though after the initial release of this paper others began to do so.⁶ Third, we introduce a complementary analysis of unit level output behavior to determine whether capacity was being strategically withheld to drive up market prices.

2. BACKGROUND

The California market institutions in place in the Summer of 2000 were introduced in April 1998 after four years of debate about electricity sector restructuring and the design and creation of complex new wholesale market institutions (Joskow, 2000). Under California's electricity restructuring and

5. The data available to us are not sufficient to measure supplier withholding behavior by generators located outside of the CAISO. Nor can we measure the control over generation supplies acquired by wholesale market aggregators or their bidding and supply behavior. Yet, as we will demonstrate, net imports into California can have significant effects on market-clearing prices. These imports declined significantly in Summer 2000 compared to Summer 1999 and wholesale marketers were likely to have been active participants as buyers and sellers in the California markets. It is possible that generators, or wholesale market aggregators, controlling supplies from generating units outside of California may also have had the incentive and ability to increase wholesale market prices in California (and the rest of the WSCC). Accordingly, a complete and definitive picture of wholesale market behavior and performance in California during this period, and the effects of strategic behavior by suppliers with market power, requires an analysis of demand and supply conditions in those portions of the WSCC that historically have provided the bulk of the net supplies to California. Such an analysis should also take account the control over generation supplies accumulated by wholesale marketers operating in the WSCC. The information necessary to perform this analysis is neither publicly available nor available to the CAISO. This is the area where additional data collection and appropriate empirical analysis by responsible regulatory agencies can provide value-added to the extensive analysis of market behavior and performance of California suppliers that has been completed over the last two years.

6. See, for example, Borenstein, Bushnell and Wolak (2002).

deregulation program, wholesale market prices were intended to be “market-based.” The non-profit CAISO was created to operate the transmission networks owned by the state’s Investor Owned Utilities (IOUs) and the PX was created to operate day-ahead hourly auction markets for wholesale electrical energy.⁷ The CAISO was also given the responsibility to operate hourly auction markets for reserves (ancillary services) and imbalance energy and to manage congestion. All supply from generators selling into the CAISO control area and all demand by “load-serving entities” located in the CAISO control area must ultimately be physically scheduled with or dispatched by CAISO.

Energy to meet California loads comes from both in-state generators and out-of-state generators. The in-state generators consist of four nuclear power plants, hydro-electric plants that are located primarily in Northern California, gas-fired steam and peaking turbines, and cogenerators and other generation sources that are “Qualifying Facilities” (QFs) under the Public Utility Regulatory Policy Act of 1978 (PURPA). About half of in-state generating capacity consists of gas-fired steam and peaking units and these units are the marginal supply sources during most hours in the Summer when electricity demand in California is highest. These units were sold by the three incumbent utilities in 1998 and 1999 to five independent power companies and these new “merchant generators” owned these units during the period we study. It is fairly easy to measure the marginal costs of these units since their thermal efficiencies at different output levels are well known and spot market prices for natural gas are available from a variety of sources. No new generating capacity had entered the California market between the time these generating units were divested and the period we study; most of this gas-fired capacity dates back to the 1960s and 1970s.

During the Summer months, the marginal supply resource that clears supply and demand is typically a conventional steam or combustion turbine unit fueled by natural gas or oil. Figure 1 depicts the marginal cost curves for this gas-fired generating capacity in CAISO’s control area, assuming that the price of gas is either \$2.50/Mcf (as in 1999) or \$6/Mcf (as in late Summer 2000). These marginal cost curves can be thought of as the “top” of the CAISO area’s competitive generation supply curve during the Summer months. During Summer hours, a competitive market would clear somewhere along these supply curves. Changes in natural gas prices shift the supply curve up or down and, other things equal, competitive market prices would move up or down along with the changes in gas prices. Changes in demand move the equilibrium competitive price along this supply curve so that competitive prices increase directly with demand. As we shall discuss, tradeable permits for NO_x emissions increase and “twist” the marginal cost curve depending on the price of NO_x.

7. The PX ceased functioning in January 2001 and subsequently filed for bankruptcy. During its existence, the PX also operated “hour-ahead” and monthly block forward markets, but they were of little quantitative or financial significance and will not be discussed further here.

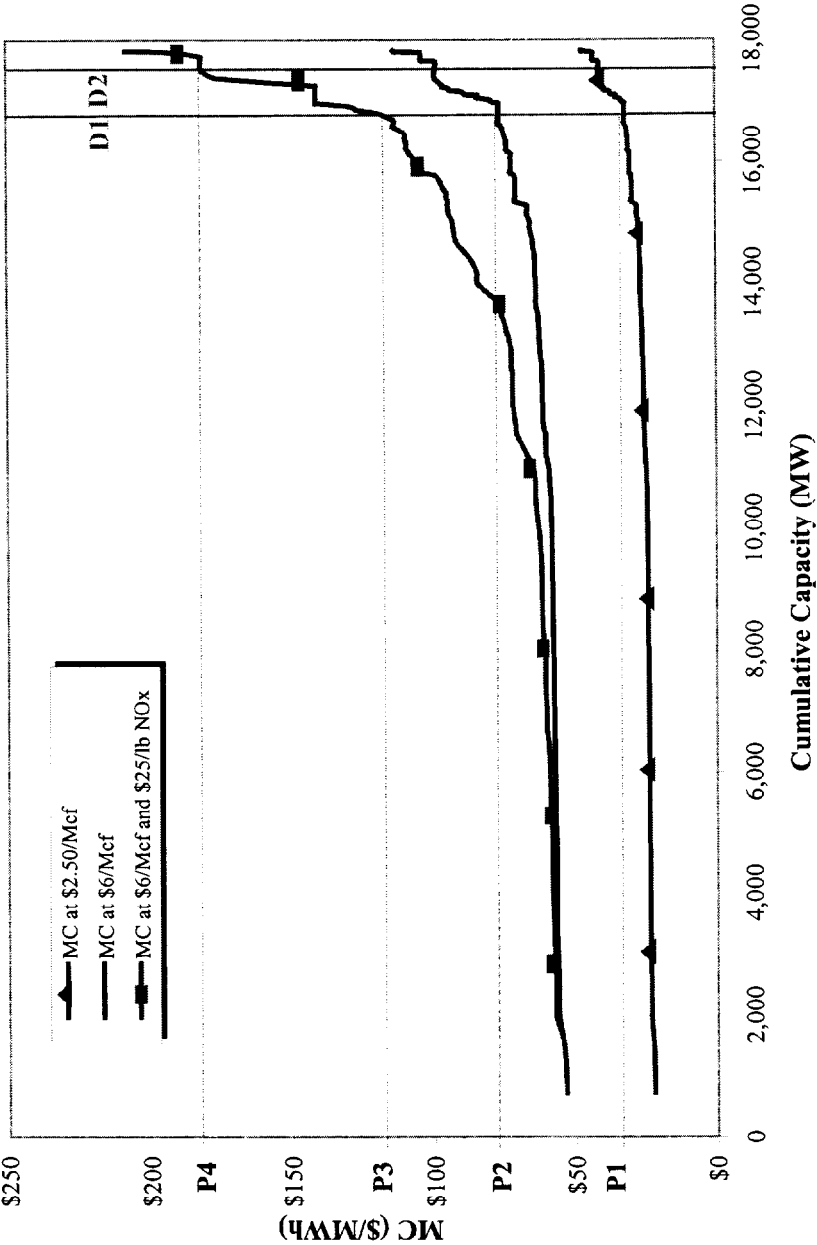
credits and differences in emissions rates across generating units, so that the competitive price for electricity increases directly with NO_x permit prices. In addition, at high demand levels, the competitive supply curve is much steeper with a NO_x permit trading system than without one. This is the case because the generating units with the highest emissions rates produce as much as 50 times more NO_x per unit of electricity output than those with the lowest emissions rates while the difference in marginal fuel costs between the most efficient and least efficient generating unit is only a factor of about two.

Until 1998, the roughly 18,000 MW of gas-fired capacity in the CAISO's control area was owned by the three vertically integrated IOUs. Under California's restructuring program, these utilities were required to sell this capacity to independent companies or New Generation Owners (NGOs). As noted above, most of this capacity was ultimately sold to five out-of-state companies with large national unregulated power plant businesses. The nuclear and hydroelectric capacity, and the high-price contracts with QFs, were retained by California's three IOUs. This amounts to about half of the original in-state generating capacity originally owned by or contracted for by these utilities prior to restructuring. The hydroelectric capacity retained by the IOUs has limited energy production capabilities over the course of the year dictated by reservoir storage capacity, water runoff, and water release constraints. California has historically imported large quantities of electricity from neighboring states and during peak demand periods imports are necessary to balance supply and demand. Understanding the role of imports in the restructured markets is a significant methodological challenge.

California's restructuring program included a "retail competition" option which permitted all retail consumers to arrange for their power supplies with an unregulated retail electricity service provider (ESP) of their choice. ESPs arrange for power supplies in the wholesale market and deliver it to consumers over one of the utility's distribution networks. Consumers who do not voluntarily choose an ESP continue to receive "default service" from one of the three IOUs as they always have. About 85% of the retail demand continued to be supplied by the utilities during 2000.

It is particularly important to note that the short run elasticity of demand for electricity in California is close to zero and is almost completely unresponsive to swings in hourly prices since few consumers have hourly recording meters or the communications and control equipment to interact directly with the wholesale market. Moreover, during the time period we study, while wholesale prices were effectively deregulated, retail prices for generation service continued to be regulated for up to four years based on a pre-determined retail price of roughly \$60/MWh.

Figure 1. Marginal Costs for Gas Units



California's restructuring and competition rules required the IOUs to serve all of their default service demand from the PX and ISO spot energy markets.⁸ They were also required to bid all of their remaining generation supplies into the PX and ISO spot markets. Independent generation suppliers and non-utility demands were not required to deal through the PX or ISO markets, but could instead enter into bilateral contracts and self-supply ancillary services. Since the utilities retained responsibility for such a large fraction of the demand, most of the wholesale trade in electricity took place either in the PX's day-ahead market or in the ISO's real-time balancing market.

Generators can receive revenues from several sources. They can sell energy to the PX and ISO. They may also enter into forward contracts with entities other than the three California IOUs. Finally, they can earn revenues by supplying "ancillary services" to the ISO. These services are reserves that the ISO can call on to manage imbalances in supply and demand and to deal with congestion. Generators selected to provide ancillary services effectively enter into an option contract with the ISO. They are paid a market-clearing price (day-ahead or hour-ahead) to hold capacity in reserve and available to the ISO. They are then paid for any energy that the ISO calls them to provide, based either on the market clearing price for energy or their bid, whichever is higher.

Our analysis of prices focuses on the hourly day-ahead unconstrained prices observed in the PX during the Summer of 2000. We focus on the PX because it was the venue where the bulk of the energy was traded. Moreover, there appears to have been reasonably efficient arbitrage between the PX market, the bilateral day-ahead market (Joskow, 2000), and the real-time market (Borenstein, Bushnell, Knittel and Wolfram, 2001).⁹ We focus on unconstrained prices (that is, pre-congestion management) for simplicity, though there was relatively little significant transmission congestion during Summer of 2000. We do take congestion into account in our analysis of supplier withholding. It should be noted, however, that our analysis ignores the ancillary services revenues earned by these suppliers and, accordingly, does not cover all of the revenues they receive from the market.¹⁰

Table 1 displays the average hourly volume-weighted prices for each month from April 1998 through December 2000. The fourth column of the table is a comparable set of forecast prices for year 2000 published by the California Energy Commission (CEC) in March 2000. The table indicates that PX prices

8. Some forward contracting was permitted by the Fall of 2000 and some limited forward transactions took place in a block forward market run by the PX during 2000 as well.

9. Joskow and Kahn (2001b) examines whether our supply withholding behavior is affected by using real-time prices rather than day-ahead prices using June as a test case. We also examine a variety of other issues raised by Harvey and Hogan (2001a) in that paper. These considerations do not affect our basic results.

10. Joskow and Kahn (2001b) explore the effects of incorporating ancillary services revenues into the analysis of the profitability of marginal generating units, an issue raised by Harvey and Hogan (2001a).

were roughly in line with expectations during 1998 and 1999 and the first four months of 2000. Beginning in May 2000 prices began to rise and then rose to unprecedented levels in June. Prices moderated somewhat in July and then jumped significantly in August before moderating a bit again in September. Prices throughout the Summer months of 2000 were four to five times higher than in 1998 and 1999 and the CEC's projections for 2000. While we have not analyzed the post-September prices, it should be clear that prices did not return to "normal" levels and exploded again in December. There are a number of unusual events that affected California's electricity markets after October that make this period difficult to analyze: an order of magnitude increase in gas prices during December, gas shortages, changes in market rules, a large quantity of plant outages, utility credit problems, and other factors. Our analysis focuses on PX market clearing prices during the May through September 2000 period.¹¹

Table 1. California PX Day-Ahead Prices (\$/MWh, Weighted Averages 7 x 24)

Month	1998	1999	2000	2000 (CEC)*
January	-	21.6	31.8	27.7
February	-	19.6	18.8	24.1
March	-	24.0	29.3	23.3
April	23.3	24.7	27.4	20.0
May	12.5	24.7	50.4	18.5
June	13.3	25.8	132.4	18.8
July	35.6	31.5	115.3	28.0
August	43.4	34.7	175.2	40.9
September	37.0	35.2	119.6	45.3
October	27.3	49.0	103.2	32.2
November	26.5	38.3	179.4	31.6
December	30.0	30.2	385.6	30.7
Average	30.0	30.0	115.0	28.5

*California Energy Commission Forecasts, 3/13/00

11. See Joskow (2001) for a discussion of the entire period. It should be noted that the prices in Table 1 do not reflect "fully unregulated" wholesale prices. Until July 2000 there was a \$750/MWh cap on prices. This cap was reduced to \$500/MWh during July and then to \$250/MWh in early August. The \$500/MWh and then \$250/MWh cap were binding during many hours in August and September. Technically, the cap was on prices in the ISO's real-time market. However, since it would have been irrational to pay more than the real-time market price cap in the day-ahead market, this became the effective cap on day-ahead prices in the PX as well. During emergency situations, it was widely known that the ISO would pay more than the price cap for supplies and this probably had the effect of creating more emergencies as generators stopped scheduling supplies day-ahead or hour-ahead in the hope of getting higher prices from the ISO through a last-minute "out of market" sale. The analysis here ignores the price cap and simulates unconstrained competitive benchmark prices. Since we are comparing these simulated prices to actual market prices which reflect the effects of price caps we are likely to underestimate the true potential price gap attributable to market power.

3. METHOD FOR ESTIMATING COMPETITIVE BENCHMARK PRICES WITH PUBLIC DATA

In this section we estimate competitive wholesale market benchmark prices and compare these benchmark prices to the prices that were actually observed. We simulate an energy market in which all demand clears in a single market, i.e., we do not attempt to simulate the relationship between day-ahead and real-time markets.¹² The more the observed price exceeds the competitive benchmark price, the more one can presume that either market power was being exercised or some other source of market imperfection has interfered with the competitive interplay of supply and demand.

The competitive price benchmark that we utilize is the short run marginal cost of supplying electricity from the last unit that clears the market in each hour. Comparing realized prices with marginal supply costs in this way is a widely accepted method for measuring the presence of market power, and is especially useful for examining prices in commodity markets with homogeneous products like spot electricity markets. We recognize that modest departures from ideal competitive conditions do not necessarily imply that there is sufficient market power to be of policy concern; many markets that are not subject to price controls are imperfectly competitive. Moreover, any empirical analysis of pricing behavior is subject to some degree of uncertainty. Finally, we recognize that prices may depart from observed marginal cost even in a perfectly competitive market to reflect real capacity constraints and opportunity costs associated with inter-temporal production limits on energy-limited generators such as hydroelectric plants. However, this approach allows us to quantify how far realized market prices depart from competitive benchmark prices and provides a useful metric, *along with our analysis of withholding behavior*, that policymakers can utilize to come to a judgment about whether the gap between competitive benchmark prices and actual prices is so large that regulatory interventions are justified.

This approach to measuring market power in wholesale electricity markets was pioneered by Wolfram (1999) in her study of the electricity market operating in England and Wales. The same approach has been applied previously in studies of the California market (Borenstein, Bushnell and Wolak, 2000¹³; Wolak, Nordhaus and Shapiro, 2000; and Hildebrandt, 2000). These studies of the California market relied on confidential CAISO data to which we do not have access. Accordingly, our empirical methods differ from these earlier studies because it must be adapted to the limitations of public data, and our goal

12. We note again that generators earn additional revenues from supplying ancillary services to the ISO. These revenues are especially important for covering the fixed costs of peaking units that supply energy infrequently but serve as operating and replacement reserves much more frequently.

13. Borenstein, Bushnell and Wolak (2002) updates and extends their earlier study in a number of ways, including an estimate of the role of NO_x credits.

of making relatively simple, but robust estimates. In addition, our work also extends the basic benchmark price approach adopted by these earlier studies by adding a complementary analysis of supplier behavior. We now describe each major element of our computational approach.

Load Slices

We are constrained to analyze months as homogeneous periods because we only have hydro data available on a monthly basis (see below). Within each month, we rely, for simplicity, on 100 demand or load periods. That is, we take the ISO's hourly demand data and segment it into 100 demand periods varying from the lowest to the highest percentile of hourly demand for the month. Within each load period, we look at the mean load in the period and use the intersection of that demand with the supply curve for the month to estimate the mean price for that load point. We add 3% to each demand level reflecting the CAISO's demand for ancillary services capacity.¹⁴

Hydro

Public data on hydroelectric output is only available on a monthly basis. EIA Form 759 gives output at the unit level. These data allow us to separate units that are dispatched by the CAISO from other California hydro units, but provide no information about how to allocate the energy from the relevant units to different time periods. We have tried assigning this energy to periods within each month using different algorithms. These algorithms assign energy to higher demand periods up to a maximum subject to the constraint that every period receive some minimum amount of hydro energy. Our base case relies on an algorithm which limits the amount of hydro energy in each period to a minimum of 60 percent of the amount that would be assigned to each hour if hydro energy were spread evenly throughout the month and a maximum of 8,500 MW. This is a conservative procedure that may tend to allocate less hydro energy to high demand periods than actually occurs, leading to higher estimates of competitive peak period prices for electricity.

The 8,500 MW hydro maximum is used by the ISO as their estimate of hydro capacity available in their control area (CAISO, 2001, p.9). It represents approximately two thirds of the hydro capacity inside of the ISO.¹⁵

14. In Joskow and Kahn (2001a) we followed Hildebrandt (2000) which includes a 10% adjustment for ancillary services, representing 3% for regulation and 7% for reserves. We have now become convinced by others, including Harvey and Hogan (2001a), that the 10% adjustment is too large and overestimates competitive benchmark prices. Accordingly, in this paper we use 3%, representing expected demand for regulation energy.

15. Based on EIA Form 860, we count just under 12,000 MW of hydro capacity inside of the ISO control area including all hydro and pump-storage capacity in California besides that owned by Los Angeles Department of Water and Power (LADWP).

Because of long-term contracts and agreements, such as those between federal agencies and many California municipal utilities, not all hydro capacity is available to meet peak demand. The 8,500 MW figure is approximately the capacity that can be dispatched by SCE and PG&E and hence is likely to be price-responsive.

Plant Outages and Availability

Forced and planned outages are different phenomena. Planned outages for maintenance are typically scheduled in low demand periods. This has the effect of equalizing reserve margins across months, to the extent possible. This is a common procedure in the industry and in production simulation modeling. During the Summer period there should be no planned maintenance, and we do not include any allowance for it. We use standard industry data (NERC, 2000) on historical average forced outage rates by unit type to adjust the marginal cost curve (i.e., shift the supply curve backwards) to reflect “non-strategic” forced outage rates. This procedure is sometimes referred to as “de-rating” the nominal capacity of units to a “firm” capacity level. The forced outage rates for the gas plants are in the 6% to 13% range.¹⁶

Wind turbine generators present a special problem. The CAISO applies an 80% unavailability factor to account for the random availability of wind power. We adopt this conservative view and convert the 1,876 MW maximum capacity of wind turbine generators into 375 MW of firm capacity.¹⁷

Our methods for reflecting forced outages differ from those of Borenstein, Bushnell and Wolak (BBW). BBW use a Monte Carlo simulation of forced outages for in-state fossil generation. BBW argue that maintenance decisions for these units are strategic variables and, therefore, they make no estimate of such outages for in-state fossil generators. By relying on settlements data for must-take resources, BBW are reflecting both maintenance and forced outages for all of this capacity. In contrast, we apply the outage treatment for in-state fossil to must-take resources as well, since we do not have hourly outage information.

Our derating procedure underestimates supply if actual outages were below the historical levels reflected in the outage data we utilized. Of course, one of the rationales for introducing competition into the electric power industry was that market incentives would lead competitive suppliers to *increase* availability, reduce forced outages, and increase effective capacity. Although it is difficult to verify actual outages for many resources, plant-level monthly energy data are available from EIA Form 900. We found that both Diablo Canyon and San Onofre nuclear power plants (which remained in the hands of

16. These data define a number of different outage rates. We use the EFOR (equivalent forced outage rate).

17. See CAISO (2001), pp. 13-15.

the incumbent utilities and were subject to complex regulatory transition arrangements) ran at full capacity over the Summer months. Accordingly, we place them in our supply curve at full capacity.

Imports

We adopt the approach of Borenstein, Bushnell and Wolak, 2000 (BBW 2000) with regard to modeling imports. They argue that high observed prices in California draw in more imports than would occur under lower competitive prices, other things equal. BBW (2000) use confidential data on adjustment bids to characterize this elasticity. We have assumed an elasticity of 0.33. This elasticity is loosely based on BBW's claim that imports would be 5.3 percent lower (p. 30) and prices approximately 15.5 percent lower (p. 33) under marginal cost pricing. Given the imprecision of their elasticity estimates,¹⁸ an elasticity of 0.33 is well within the range of what they find. We then use data on observed net imports, and PX prices to impute net imports under marginal cost pricing. In other words, for each period and for every price level c , we calculate the amount of infra-marginal net imports as follows:

$$netimp(c) = \left(\frac{c}{p_{px}} \right)^{\eta} * netimp(p_{px})$$

where η is the elasticity of net imports, p_{px} is the realized PX price, and $netimp(p_{px})$ is the realized level of net imports at the realized PX price. Our benchmark price is then the c at which the sum of estimated net imports and infra-marginal in-state generation, including must-take generation, clears the market.

We rely upon imports to clear the market when in-state fossil supply is exhausted. Because this will occasionally require more net imports than what was actually observed, our procedure will raise their price substantially when this is required.¹⁹ These prices will be higher than the ISO price caps in place during the Summer. We interpret these cases as corresponding to the ISO's purchase of Out of Market (OOM) energy.

In-State Fossil Generation

Natural gas costs for weekdays at the Southern California burnertip and at Malin were provided to us by Southern California Edison, from trade

18. BBW's estimate of 5.3 percent has a standard error of 8.1 percent.

19. Our measure of imports follows BBW (2000). In particular, we assume that SCE's share of Mohave is included as inside generation, and not net imports. This follows the Joskow and Kahn (2001b) correction of Joskow and Kahn (2001a).

publications. We add transport costs to the Malin prices to bring the costs to the burnertip in Northern California. Because we are constrained to a monthly level of analysis, we use monthly averages of these prices. The monthly gas price values used are given in Section 4, where we present results.

We rely upon the Henwood Energy Services Incorporated (HESI) commercially available database for the WSCC to characterize heat rates, and variable O&M costs for in-state fossil generators. The heat rate data are consistent with those found in Klein (1998). There are many definitions of unit capacity, which result in quantitative differences that typically are small.²⁰ Accordingly, we adopt for this analysis the capacities posted on the ISO website.²¹

RECLAIM NO_x RTC Prices

One factor that can affect competitive market prices for electricity which neither BBW (2000) nor the CAISO addresses involves the impact of the air emissions regulatory framework in California. California has extremely stringent air quality regulations. One pollutant of particular concern is nitrogen oxide (NO_x). As explained above, regulation of emissions in the Los Angeles area is controlled by the SCAQMD, which operates the RECLAIM Trading Credit (RTC) emissions permit trading program for NO_x emissions from electric generating units and other stationary sources. Under this program NO_x emissions are regularly reported during pre-established cycle periods of one-year's duration. The owners of a source of NO_x emissions must reconcile NO_x RTC allowances with reported emissions within 60 days of the end of the 12-month reporting period.²² The RTC program resembles the SO₂ permit trading regime authorized under the 1990 Clean Air Act Amendments.²³

We would expect competitive generation suppliers to include the prices of RTC NO_x credits in their bids even if these credits had been previously acquired at much lower prices (or for free). This is the case because these emissions credits could be sold to other affected sources at their market value and thus represent a legitimate competitive market opportunity cost. RTC allowances had been selling at very low prices (\$1-2/pound) through the early part of 2000. Since most generation in the SCAQMD area emits 1 lb/MWh of NO_x or less, emissions costs internalized into electricity prices would be \$1-2/MWh at most during this period. Starting in Spring 2000, however, RTC

20. Harvey and Hogan (2001a) took issue with the definition used in Joskow and Kahn (2001a). Joskow and Kahn (2001b) use different sources.

21. See <http://www1.caiso.com/docs/2001/04/02/2001040211441714244.xls>

22. NO_x permits are issued to cover emissions during a particular 12-month period or cycle. Because the emissions sources are divided into two groups with cycles that overlap by six months, the effective cycle is 18 months.

23. The SO₂ emissions trading program is described in detail by Ellerman et al. (2000).

prices began to increase substantially.²⁴ By June they were nearly \$10/pound. This would add \$10/MWh to MCP most of the time, and much more when gas turbines with much higher emissions rates, e.g., some turbines emit in excess of 6 lb of NO_x per MWh, are producing electricity. NO_x RTC prices continued to climb throughout the Summer, rising to around \$35/pound by late August. At these levels, NO_x RTC requirements significantly affect price during all hours for which fossil plants in the SCAQMD clear the market, but especially during peak periods when gas turbines are on the margin. Therefore, we decided to add the effects of NO_x RTC prices to our estimates.

For most units in SCAQMD that were formerly owned by SCE, we rely on estimates of NO_x emissions rates provided to us by SCE based on publicly available data and regulatory filings. For other units, we rely on NO_x emissions rates from the HESI databases.

4. RESULTS

Table 2 below presents our estimates of competitive benchmark prices for May through September 2000. We report a range of prices, reflecting alternative assumptions about NO_x RTC prices. Table 2 also displays the actual average day-ahead PX prices during these months of year 2000 for comparison purposes. The data on NO_x RTC prices are difficult to interpret for a variety of reasons. There is general agreement that NO_x RTC prices were increasing between May and September. Finding an appropriate price for each month requires that we interpret the data from SCAQMD carefully. We give a full discussion of the choices we have made in the Appendix. Table 2 indicates in bold the benchmark wholesale market price associated with our choice of the most appropriate NO_x RTC price for each month.

Table 2. Competitive Counterfactual at Different RTC Costs (2000)

Month	Average PX Price (\$/MWh)	Competitive Benchmark Price (\$/MWh)					Average Gas Price	
		Assumed NO _x Price					(\$/MMBtu)	
		\$0/lb	\$10/lb	\$20/lb	\$30/lb	\$35/lb	North	South
May	47.23	55.11	58.56	61.79	64.66	64.63	3.77	4.11
June	120.20	64.84	67.23	70.14	73.38	74.99	4.59	4.99
July	105.72	58.62	60.91	63.25	65.60	66.72	4.35	4.97
August	166.24	86.96	92.02	96.97	102.40	105.15	4.84	5.69
September	114.87	74.08	78.34	83.07	86.88	88.96	5.88	6.64

24. We have not analyzed why NO_x RTC credit prices increased so much during the summer of 2000 or whether the observed price increases are consistent with competitive behavior in the RTC credit market. A careful analysis of behavior and performance of the RTC credit market would also be a worthwhile undertaking.

It is clear from Table 2 that there is a significant gap between the competitive benchmark prices that we estimate and actual market prices in June, July, August and September 2000. We want to emphasize that this gap between competitive benchmark prices and actual market prices takes into account the effects of gas prices, demand levels, import levels, and NO_x credit prices; the "market fundamentals" that have often been identified as contributing to higher prices in Summer 2000 than in Summer 1999. It is also interesting to note that if NO_x credit prices had remained at 1999 levels, competitive benchmark prices would have been reduced significantly, especially in August. We believe that the estimated price gap is large enough to provide compelling evidence that market power or other market imperfections lead to a significant increase in prices above competitive levels during Summer 2000.

Our estimates of competitive benchmark prices are roughly similar to those obtained in other studies using similar techniques and confidential data to which we do not have access. BBW (2002) does include the cost of RTC allowances in their estimates. Nonetheless their estimates of competitive benchmark prices are lower than ours for the same period. Some of this difference is due to their use of average NO_x credit prices (BBW 2002, p.41), whereas we use an estimate of higher marginal prices. This choice does not, however, explain all of the differences.

Import Sensitivity

One use of the framework that we have applied to develop competitive benchmark prices is to examine hypotheses about the effects of key variables on competitive market prices. Here we examine the effect on wholesale prices of reductions in net imports between 1999 and 2000. Many commentators have remarked on the significant decline in net imports between Summer 1999 and Summer 2000. Table 3 shows the mean difference in actual monthly net imports from year to year. It also compares the estimated benchmark price before considering effects of RTC credit prices (i.e., based on the zero RTC credit price column in Table 2) with our estimate of what the competitive benchmark price would have been if the 1999 level of imports had occurred. It is clear that prices are higher in Summer 2000 as a result of lower net imports, but if NO_x emissions were not an issue, the impact of reduced imports alone accounts for a relatively small fraction of the actual increase in wholesale prices from Summer 1999 to Summer 2000. As NO_x RTC prices rise toward the end of the Summer, the reduced level of imports becomes a much more important factor in explaining wholesale price increases. This is the case because as imports fall, in-state generating units with relatively high emissions rates run more often to balance supply and demand.

Table 3. Net Import Sensitivity

		June	July	August	September
1999 actual average hourly net imports (MWh)		5,871	6,633	6,539	7,070
2000 actual average hourly net imports (MWh)		4,262	3,621	3,162	4,386
MCP with 1999 net imports (\$)	NO _x \$0/lb	60.33	47.10	58.08	64.89
MCP with 2000 net imports (\$)		64.84	58.62	86.96	74.08
MCP with 1999 net imports (\$)	NO _x \$10/lb	62.22	48.47	62.18	67.30
MCP with 2000 net imports (\$)		67.23	60.91	92.02	78.34
MCP with 1999 net imports (\$)	NO _x \$20/lb	65.08	49.56	65.02	69.17
MCP with 2000 net imports (\$)		70.14	63.25	96.97	83.07
MCP with 1999 net imports (\$)	NO _x \$30/lb	68.05	50.46	68.19	70.75
MCP with 2000 net imports (\$)		73.38	65.60	102.40	86.88
MCP with 1999 net imports (\$)	NO _x \$35/lb	68.52	50.95	68.18	71.35
MCP with 2000 net imports (\$)		74.99	66.72	105.15	88.96

5. WITHHOLDING AND UNILATERAL MARKET POWER: THE ECONOMIC LOGIC

The previous analyses shows that “market fundamentals” cannot fully account for the high levels of observed prices in the Summer of 2000. Even after accounting for lower levels of imports and very high NO_x RTC prices, we still observe a large deviation of wholesale market prices from the competitive benchmark price, i.e., marginal costs of supplying additional electricity at the associated market clearing quantities. However, while we observe large price/marginal cost margins during the Summer of 2000 which we believe are inconsistent with competitive markets, our analysis so far does not measure behavior that is likely to be the cause of these high prices. It has been conjectured, for example, that the high observed prices simply reflect scarcity rents that arise when demand is high, capacity constraints are binding, and competitive market prices must rise to clear the market (CaPX, 2000). In the next two sections we investigate the hypothesis that withholding behavior by generators in California is one cause of the large measured gap between prices and marginal costs. It is clear from first principles that supply withholding could be the source of high prices. Whether this is, in fact, the case is an empirical question.

We begin by presenting a simple example to demonstrate the unilateral profit maximization logic behind capacity withdrawal and show that rational capacity withholding does not require collusion among suppliers. We consider the unilateral case, i.e., only one portfolio player adopts this strategy and all other generators behave competitively and bid at prices equal to their marginal cost. We can characterize the profit effects of capacity withdrawal simply as the

sum of two effects. These are (1) the increased profits on the capacity offered after withdrawal due to the ability to raise price, and (2) the lost profits of capacity withdrawn (see Wolfram, 1998). The profit changes must take account of the cost reduction due to not producing from the withdrawn capacity. We can express these effects as follows:

$$\Delta \text{Profit} = \Delta \text{Price} * \text{Remaining Quantity} - \Delta \text{Capacity} * \text{Competitive Price} + \Delta \text{Operating Cost}$$

This expression is derived formally in Joskow and Kahn (2001a).

As is apparent from the formula, whether withdrawing capacity is in the self-interest of a portfolio generator will depend critically upon the slope of the supply curve. It must be steep enough to result in market clearing prices (MCP) sufficiently high so that the increase in profit on generation still tendered to the market more than offsets the profits lost on the capacity withdrawn. We construct some examples based on our benchmark estimate of competitive supply conditions that prevailed in June 2000. These estimates come from our simulations (with NO_x effects) for June 2000. We examine one case, at higher loads, where a load increase or capacity withdrawal of 1,500 MW results in a price increase of about \$36 per MWh. Our second case, at lower loads, produces only a \$3 price increase for the same 1,500 MW withdrawal. The method for estimating the price increases from the supply curve is illustrated in Joskow and Kahn (2001a).

Table 4 displays the profitability effects of capacity withholding for the two examples.²⁵ In both cases, we make three assumptions: (1) the portfolio generator with 3,000 MW of capacity produces only half that amount, (2) the competitive MCP is \$60/MWh, and (3) the generator's marginal cost is \$55/MWh. In the first case, where capacity withdrawal raises price significantly, the revenue gain is large. In the second case, the impact of withholding capacity on price is relatively small because supply is much more elastic over the relevant range of output. Withholding is unprofitable since the increase in price on the tendered generation does not offset the lost profits on generation withheld. The details of these results are shown in the following table.

Table 4. Unilateral Market Power Examples

Case	Revenue Loss	Revenue Gain	Cost Savings	Δ Profit
Δ Case 1 High Price Increase	90,000	54,000	78,750	42,750
Δ Case 2 Low Price Increase	90,000	4,500	82,500	-3,000

25. This example differs slightly from a similar calculation in Joskow and Kahn (2001a). The changes reflect re-estimation of the supply curve.

These stylized examples are constructed to make it difficult to find unilateral market power. They rely on the assumption that only one supplier withholds capacity, while all of the other suppliers behave competitively. In the California electricity market during the Summer of 2000, it appears that more than one portfolio player was implementing a withholding strategy. We illustrate this claim empirically in the next section. If multiple suppliers are behaving strategically, the effects of withholding on wholesale market prices could be much larger than suggested by these simple examples.

6. EMPIRICAL ANALYSIS OF WITHHOLDING

Now we turn to the analysis of physical supply and withholding behavior. We use plant and unit level output data from EPA, the ISO's real-time dispatch, and the Western System Co-ordinating Council (WSCC) to examine the physical behavior of the price setting firms to determine whether there was really "scarcity," that is, that demand was so high that competitive prices above marginal cost were necessary to clear the market, or whether generators withheld supplies from the market when it would have been profitable for a generator without market power to supply more.

We restrict our analysis to a set of high-priced hours when it should have been economical for virtually all of the fossil generators to supply, absent market power. In particular, we look at hours when the real-time price²⁶ was greater than 17,000 Btu/kWh times the delivered gas price plus 1 lb NO_x/MWh times the monthly RTC price. The heat rate threshold covers virtually all steam and most peaking units. The NO_x emission rate covers almost all steam units. Units with higher costs should be reserved to provide ancillary services. Table 5 shows the average price and number of hours per month that meet this criterion.

Table 5. Monthly Cut-Off Prices and Number of High-Price Hours

Month	Monthly Cut-Off Averages (\$/MWh)		Number of High-Price Hours	
	SP15	NP15	All Hours	Hours without South to North Congestion
June	95	89	104	96
July	105	95	124	114
August	132	118	271	241
September	148	139	82	66

26. We focus on the real-time price, as opposed to our first analysis (2001a) that used day-ahead prices. Harvey and Hogan (2001a) correctly observe that the real-time price is a better indicator for production data since it reflects all output decisions by suppliers. As we observed earlier, however, there is a close correlation between these prices.

We compare observed levels of production by units likely to be setting prices, with their maximum generating capacities during those hours.²⁷ Northern California (NP15) generation is analyzed separately from Southern California (SP15). There is a substantial “output gap” between observed and maximum possible levels of generation in both zones. Three non-strategic factors may explain this gap: (1) capacity may be covering the CAISO’s operating reserve or “ancillary services” requirements, (2) capacity may be out of service due to forced outages, and (3) interzonal transmission constraints (South to North) may limit economic dispatch of SP15 plants. Therefore, we test whether the gap can be explained by these three factors. If the gap cannot be explained, we conclude that it is indicative of generator withholding resulting either from high bids that do not clear the day-ahead or real-time energy markets or direct withholding of capacity from these markets.

- The ancillary services tests have two elements: (1) we compare the zonal CAISO ancillary services requirement in the selected hours against the output gap, and (2) we consider whether the CAISO dispatched reserves during our sample hours. If reserves were dispatched, they will appear as production in our data and would, therefore, not explain any output gap.
- The forced outage test is necessarily limited in its applicability. We apply three outage tests to the data to ascertain whether forced outages might explain output gaps.
- The congestion test requires that we identify congestion during our sample hours. Production levels in hours without a constraint should not be affected by transmission issues.

We rely on three data sources for this analysis. Each has hourly production data, but the sources differ by the units covered. EPA’s Continuous Emissions Monitoring System (CEMS) database tracks hourly production and emissions of certain pollutants that are regulated under the Clean Air Act. CEMS data are available on the EPA’s website. CEMS data do not include gas turbines (GT) and some small thermal units. Table 6 below lists the largest units that are omitted from the CEMS database, their ownership, capacity and NO_x emission rates. About 1,300 MW of gas capacity is excluded from the EPA data. We address units excluded from CEMS in two ways. For most peaking

27. These calculations correct for the Daylight Savings Time issue identified in Harvey and Hogan (2001b).

units, we rely on the ISO BEEP stack dispatch.²⁸ The BEEP data record the energy dispatched from units in real time, but do not include any energy that may have been scheduled before real time. Because we are using BEEP to characterize the output of GTs and GTs generally run fully loaded, when we observe output for a unit in a given hour in BEEP we assume that the unit operated at full load for that hour. The third source gives data at the plant level, not the unit level. These data are from the WSCC's Extra High Voltage (EHV) database. The EHV data, which are available to all WSCC members, were provided to us by SCE. We rely on these data only for the Long Beach units. These are rather inefficient combined cycle units, with heat rates of approximately 10,500 Btu/kWh, which could nonetheless be expected to produce energy during high-price hours.

Table 6. Units Excluded from CEMS Database

Unit	Owner	June Capacity (MW)	NO _x (lbs/MWh)
Long Beach 8	Dynegy	276	1.2
Long Beach 9	Dynegy	276	1.2
Highgrove 1-4*	Thermo Ecotek	154	1.2-2.4
Etiwanda GT	Reliant	141	5.4
Alamitos GT	AES	134	6.5
Huntington Beach GT	AES	133	5.7
Elwood GT	Reliant	48	5.4
Mandalay GT	Reliant	132	5.7

*While Highgrove reports emissions data as part of the EPA program, these plants did not operate in the relevant time period.

NO_x emission rates are calculated from publicly-available data.

Analysis of June

We begin by focusing in detail on the month of June. Our analysis attempts to discover if there is an unexplained gap between generators' capacity and observed production. An otherwise unexplained gap would tend to support our hypothesis that production was withheld by generators in an attempt to drive up price during these periods. Generating units in California typically come back into service from their annual maintenance outages during May and early June and should be ready to operate reliably through the peak Summer months. Accordingly, we would expect generating units to exhibit low forced outage rates during June even if they are run hard during that month. It is also widely accepted, we believe, that the subsequent run-up in natural gas prices, the large

28. BEEP is an acronym for Balancing Energy and Ex-Post Pricing. This software records the instructions given by the ISO to units that it dispatches in real time. BEEP stack dispatch data is available on the ISO website.

increase in demand, and the large increase in NO_x credit prices were not anticipated in June 2000 and could not have been factored into competitive supplier behavior during that month. Therefore, these factors which Harvey and Hogan (2001a) emphasize as potential explanations of low output are less relevant in June than they might be in later months. Thus, a finding that there was significant withholding of generating capacity during June 2000 is especially strong evidence supporting the exercise of market power as its source.

Logically, congestion on the transmission system could help to explain any output gaps we might identify. We therefore have examined data on interzonal transmission levels and congestion to determine the extent of transmission congestion. Since it appears that withholding was most likely in the SP15 zone, we have reviewed the possible impacts of South to North congestion on our findings. Table 5 shows that while interzonal transmission constraints do occur occasionally, they are typically limited to about ten percent of the hours with real-time prices greater than our threshold prices. In the interest of simplification, we have omitted such hours from further analysis.

We begin by computing the aggregate output gap for all five merchant generators for June hours when the real-time price was above our threshold, which averages \$95/MWh in SP15 or \$89/MWh in NP15; there were 96 hours that meet this criterion and where there was no real-time South to North congestion. In each of the 96 hours when the price was above these thresholds, we observe the hourly output of generators owned by Duke, Mirant, AES, Dynegy and Reliant. We expect to find that production in these hours will be at maximum levels. Next, we sum up the hourly output for each firm in NP15 and in SP15 separately. We compare the June production of each generator over each of the 96 hours to our estimate of their capacity. We define the output gap for each firm to be the mean difference between total capacity and the observed output in each of the 96 uncongested high-priced hours. It is important to note that the dispatch of these generators may be controlled by contractual arrangements with third parties other than the owners of the generating plants. It has been widely reported that this is the case for the AES units, which operate under a tolling agreement with Williams, but we do not know whether or how much control has been ceded to marketers through contracts otherwise. Accordingly, we use the owners simply to identify the generating plants examined and any apparent withholding observed.

Next, we want to see how much of the gap can be explained by the CAISO's reservation and use of capacity for ancillary services (we include Up Regulation, Spin, Non-Spin and Replacement Reserves).²⁹ Public data on CAISO demands for Ancillary Services (AS) are available by zone. We compare the output gap by zone to AS capacity by zone for each hour. We then check the

29. We exclude Down Regulation, because that does not require that capacity be held in reserve.

BEEP data to determine what fraction of reserves were dispatched.³⁰ Table 7 below summarizes our results.

Table 7. Mean Level of the Output Gap: June 2000

Zone	Owner	Mean Values (MWh)			
		Output	Capacity	Gap	Undispatched AS
NP15	Duke	1,469	1,485	16	
	Mirant	2,063	2,629	565	
	NP15 Total	3,532	4,114	581	1,222
SF	Mirant	206	369	163	
	SF Total	206	369	163	24
SP15	AES/Williams	2,735	3,967	1,232	
	Duke	675	717	42	
	Dynegy	1,492	2,834	1,342	
	Reliant	2,492	3,790	1,298	
	SP15 Total	7,394	11,308	3,913	1,326
ZP26	Duke	990	1,021	31	
	ZP26 Total	990	1,021	31	31

For SP15, the mean of the output gap is 3,913 MW compared to 1,326 MW for the mean of the undispatched AS demands in the zone. This leaves an average unexplained mean output gap of nearly 2,600 MW during the 96 hours, making the extremely conservative assumption that all of AS capacity requirements were covered by these units. This is 23% of the capacity of the SP15 merchant generators. That would be a very large outage rate if it were not strategic withholding. We address the difficult subject of “true” forced outages below. It is important to recognize that withholding 2,600 MW from the market during high demand conditions can have a very large effect on market prices. Referring back to Figure 1, it can be seen that a modest 1,000 MW increase in demand or reduction in supply can increase marginal supply costs by over 50% at relatively high demand levels. Based on these results, it looks as if a significant amount of capacity was being withdrawn on average during these high-price periods in SP15. Accordingly, the gap between prices and marginal costs cannot be explained by scarcity of capacity to balance supply and demand at a price equal to marginal cost. The results for NP15 are different. Here the

30. The BEEP stack is the ISO’s real-time supply curve. It consists of both bids for imbalance energy and the energy portions of bids to provide ancillary services. The BEEP data that we use to measure the output of GTs also indicate whether a unit was dispatched because an imbalance energy bid was called or because an energy bid associated with a specific ancillary service was called. So, the BEEP data can be aggregated to calculate a measure of dispatched ancillary services.

mean of the output gap is 581 MW, which is less than the mean undischarged AS capacity, 1,222 MW. Therefore, we cannot conclude definitely that there was capacity withholding in NP15.

It is important to point out, however, that this assessment is quite crude and supplies an upper bound on AS capacity requirements that might explain the output gap. This is due to two factors: (1) it neglects the possibility that hydro capacity or imports are supplying some of the AS demand, and (2) ramp rate restrictions might have made it physically impossible for the plants to supply the full AS requirement. While we have accounted for the effects of dispatching reserves on our analysis, we are unable to test the effects of alternative suppliers of AS services or those of ramp rate restrictions because of data limitations. Obviously, to the extent that some of the AS demand is being satisfied by hydroelectric capacity and out-of-state resources, as is likely to be the case, the gap would be larger by an equivalent amount.

It is interesting to note that there is no evidence that Duke was withholding output in either SP15 or NP15. Duke Energy, which appears to have been fully contracted in forward markets for 90% of its potential output, behaved much differently from Reliant, Dynegy, Mirant, and AES/Williams. Duke's production in SP15 was proportionally higher than that of these other firms. It reported much lower forced outage rates than what the other firms appeared to claim. We believe that the outage rates and production levels reflect economic incentives. If generators are not contracted, their incentive is to withhold capacity and raise spot market prices. Since 90% of Duke's capacity was contracted forward at specified prices, it had no incentive to withhold output to drive up spot market prices, and this lack of incentives appears to be reflected in its behavior.³¹

Extension to July, August and September

Table 8 extends this analysis to the months of July, August and September for the SP15 zone. The pattern of results for NP15 is not materially different in these months than in June, so we drop further discussion of NP15 since nothing can be concluded on the basis of publicly available data.

31. Harvey and Hogan (2001b) argue that AES was "the company with the highest level of forward sales" and it experienced "unusually high forced outage rates during 2000" (p.77). They are referring to the tolling agreement between AES and Williams. This is not the type of contractual arrangement that mitigates incentives to withhold output to raise prices. This tolling agreement was essentially a contract to "rent" AES' generation capacity to Williams, not a commitment by AES to supply specific production quantities at a fixed price. Under this kind of agreement Williams, not AES, was free to determine how much energy was supplied by these units and could profit if market prices increased during the summer months. As such, Williams had an incentive to withhold, consistent with the settlement that they entered into with FERC involving alleged withholding in April and May 2000 (FERC, 2001).

Table 8. SP15 Mean Output Gaps

Month	# of High Price Hours	Owner	Mean Values (MWh)			Dispatched Ancillary Services (MWh)			
			Output	Capacity	Gap	AS Total	Replacement	Spin	Non-Spin Total
June	96	AES/Williams	2,735	3,967	1,232				
		Duke	675	717	42				
		Dynegy	1,492	2,834	1,342				
		Reliant	2,492	3,790	1,298	1,756	330	30	69 430
		Total	7,394	11,308	3,913	1,326			
July	114	AES/Williams	2,757	3,967	1,210				
		Duke	635	717	82				
		Dynegy	1,811	2,765	954				
		Reliant	2,872	3,790	918	1,169	100	23	42 165
		Total	8,074	11,238	3,164	1,004			
August	241	AES/Williams	2,781	3,967	1,186				
		Duke	622	717	95				
		Dynegy	2,043	2,827	784				
		Reliant	3,076	3,790	714	1,532	183	101	79 363
		Total	8,521	11,301	2,779	1,168			
September*	66	AES/Williams	2,244	3,967	1,723				
		Duke	560	717	157				
		Dynegy	1,894	2,815	921				
		Reliant	3,072	3,790	718	1,135	130	38	39 207
		Total	7,770	11,289	3,519	928			

*Analysis for September includes days 1 through 20. The EHV data from which the Long Beach data are sourced end at September 20. Undispatch AS is shown in bold in the AS Total column. It is the difference between Total AS (in regular type) and the Total from the Dispatched Ancillary Services panel on the right side of the table.

Table 8 shows somewhat smaller output gaps in July, August and September. Net of dispatched AS, the unexplained gap for July is about 2,150 MW. In August it drops to about 1,600 MW. In September it rises again to about 2,600 MW. We can express the unexplained gaps as some kind of "outage rate," i.e., normalize them to total capacity. This calculation results in an average outage rate of between 15% (August) and 23% (June). Such rates are very high in comparison to historical average values for similar plants. The data used in our benchmark price analysis, for example, averages 7.5% and Duke's units appear to have achieved similarly low outage rates consistent with the historical experience for these generating units.

Thus far our calculations make no attempt to assess whether the output gap can be explained by unscheduled outages.³² We examine this question next. Evaluating the effects of unscheduled (forced) outages is not completely straightforward, because of the discretionary element in outages. Therefore we apply three different tests for forced outages. Test 1 measures the capacity of a generation portfolio by looking only at units that were producing any output in the hour in question. This is the strictest definition of a "no outage condition." Test 2 measures the capacity of a generation portfolio by looking only at units that were producing any output in the day in question. Finally Test 3 measures the capacity of a generation portfolio by looking only at units that were producing any output in the day in question or the day before. Another way of describing Test 3 is that an outage is real only if it occurred both the day before the day of our scarcity hours as well as the day of such an event.

Test 1 can be thought of as measuring either the withholding of a unit that could produce more in the given hour, or the occurrence of a "partial outage" in that hour. Table 9 shows that about 1,100 to 1,800 MW was not running during the high-price hours in June-September. Undispatched AS could explain some of this amount. Tests 2 and 3 employ different measures of capacity that might have run during the high-priced hours. The intuition here is that often units that might be experiencing some operating problems can be kept

32. Whether an outage is "scheduled" does not mean that it is not the result of a strategic decision to withhold output to drive up prices. The discussion of strategic behavior in electricity markets has distinguished between "physical withholding" and "economic withholding." Economically they are equivalent. When a firm seeks to affect price by simply not making some capacity available to the market it is engaged in "physical withholding." When a firm decides instead to make the capacity available to the market at a supra-competitive price, knowing that some of the capacity offered will not be selected in the associated auction process, it is engaged in "economic withholding." A supplier that chooses not to make capacity available to the market will generally declare the capacity to be "unavailable." This decision may be made well in advance of actual operations ("scheduled outage") or closer actual operations ("unscheduled outage"). Precisely how a supplier chooses to withhold, and for what reasons, is not verifiable and under the CAISO rules there are no penalties against suppliers for being "unavailable" due to either scheduled or unscheduled outages. Nevertheless, we believe that unscheduled outages are even more compelling indications of strategic behavior than are scheduled outages.

Table 9. SP15 Mean Output Gaps by Outage Test

Month	Owner	Test 1			Test 2			Test 3			Undispatched AS
		Output	Capacity	Gap	Output	Capacity	Gap	Output	Capacity	Gap	
June	AES/Williams	2,735	3,030	296	2,735	3,299	565	2,735	3,390	656	1,326
	Duke	675	693	18	675	711	36	675	712	37	
	Dynegy	1,492	2,183	691	1,492	2,386	895	1,492	2,505	1,013	
	Reliant	2,492	3,258	766	2,492	3,511	1,019	2,492	3,592	1,100	
	Total	7,394	9,164	1,770	7,394	9,908	2,514	7,394	10,200	2,806	
July	AES/Williams	2,757	3,016	259	2,757	3,249	492	2,757	3,312	556	1,004
	Duke	635	690	56	635	699	64	635	716	82	
	Dynegy	1,811	2,404	593	1,811	2,557	746	1,811	2,665	854	
	Reliant	2,872	3,272	400	2,872	3,395	523	2,872	3,518	646	
	Total	8,074	9,383	1,308	8,074	9,900	1,825	8,074	10,212	2,137	
August	AES/Williams	2,781	2,919	139	2,781	2,999	218	2,781	3,102	321	1,168
	Duke	622	701	79	622	717	95	622	717	95	
	Dynegy	2,043	2,608	565	2,043	2,760	717	2,043	2,811	768	
	Reliant	3,076	3,410	334	3,076	3,565	489	3,076	3,633	557	
	Total	8,521	9,639	1,118	8,521	10,040	1,519	8,521	10,263	1,742	
September*	AES/Williams	2,244	2,425	181	2,244	2,522	278	2,244	2,640	396	928
	Duke	560	688	129	560	699	139	560	703	143	
	Dynegy	1,894	2,479	585	1,894	2,619	725	1,894	2,649	755	
	Reliant	3,072	3,513	441	3,072	3,664	592	3,072	3,695	624	
	Total	7,770	9,106	1,336	7,770	9,504	1,734	7,770	9,687	1,917	

*Analysis for September includes days 1 through 20. The EHV data from which the Long Beach data are sourced end at September 20.

on line by operators who are strongly motivated to produce. Alternatively, if there is an economic incentive to withhold, then operators might turn them off. In such cases, “conservative” operation is also profit-maximizing.

The data in Table 9 are unadjusted for the effect of price caps on the economics of plants in SCAQMD with high NO_x emission rates. Three of the gas turbines listed in Table 6 are in SCAQMD (Alamitos, Etiwanda and Huntington Beach). With emission rates greater than 4.5 lbs/MWh, these units would have RTC costs greater than \$157/MWh in August and September when RTC prices were at \$35/lb. The fuel costs of the gas turbines would be at or above \$100/MWh during this period as well. When the price cap was lowered to \$250/MWh on August 7, these units had marginal costs above the cap. Therefore their capacity, about 400 MW total, should perhaps be excluded from the output gap estimates in Table 9 for August and September. There may also be a related issue for units with NO_x emission rates that are in the 2 lb/MWh range. At \$35/lb, these units would have \$70/MWh marginal costs for RTC credits. At the gas prices prevailing in August and September, some of these units might have marginal costs above the cut-off level for the hours that we examine. On the other hand, even these high cost units may have sold output under “Out of Market” arrangements with the ISO. We have not tested precisely the extent to which cost considerations could account for the output gap in August and September. These issues do not arise in June and July when RTC prices were lower.³³

7. CONCLUSIONS

It is clear that increases in gas prices, increased demand, reduced availability of power imports, and higher prices for emissions permits contributed to significantly higher wholesale market prices in California during 2000, compared to the previous two years. However, based on our analysis of available data, we conclude that wholesale electricity prices in California far exceeded competitive levels during June, July, August, and September of 2000. The high wholesale electricity prices observed in Summer 2000 cannot be fully explained as the natural outcome of “market fundamentals” in a competitive market since there is a very significant gap between actual market prices and competitive benchmark prices that take account of these market fundamentals.

33. Other “profitability” issues were raised by Harvey and Hogan (2001a), specifically in the context of June. Cardell (2001) raises such issues for later periods in the California market, when gas prices were substantially higher than during the summer period that we examine. We showed in Joskow and Kahn (2001b) that these issues were minimal in June. Harvey and Hogan (2001b) revisits them again arguing generally that all units which ran, or “should” have run, must be profitable ex post. This argument ignores the market uncertainties identified by these same authors. No bidder in any market characterized by uncertainty ex ante can be guaranteed profitability ex post. Harvey and Hogan (2001b) also raise specific issues about how profits should be estimated which are unsupported by any empirical analysis.

Moreover, there is considerable empirical evidence to support a presumption that the high prices experienced in the Summer of 2000 reflect the withholding of supplies from the market by suppliers (generators or marketers). We base these conclusions on results of the two analyses described herein:

- **Competitive Benchmark Price Analysis:** Observed prices in California in Summer 2000 were greater than benchmark competitive price levels. These differences are not fully explained by higher loads, reduced levels of imports, high gas prices or by high prices for NO_x RTCs.
- **Capacity Withholding Analysis:** The information that we have available to us suggests that withholding of capacity in SP15 to drive up price occurred during Summer 2000. We find a substantial gap between maximum possible levels of generation and observed levels in those hours identified as economical for all in-state generation. This gap cannot be explained by the CAISO's requirements for ancillary services or by reasonable estimates of forced outages. While our analysis of withholding is necessarily limited by the data available to us, there is sufficient empirical evidence to suggest that the high observed prices reflect suppliers exercising market power.

These empirical findings are further reinforced by the fact that the attributes of this electricity market make it likely theoretically that individual suppliers will find it profitable unilaterally to withhold output compared to price takers in order to raise market prices. In addition we found that Duke, which appears to have entered into forward contracts that eliminated or substantially reduced its incentives to withhold output, did not exhibit any withholding behavior during Summer 2000. Just as the other suppliers acted on their unilateral incentives and withheld output, Duke acted on its unilateral incentives and did not withhold output. Thus, the empirical evidence is consistent with general theoretical expectations.

Long before the new competitive wholesale electricity markets began operating in California, it was widely recognized that supplier market power could be a problem in deregulated electricity markets in general (Joskow and Schmalensee, 1983; Joskow, 1997) and in California in particular (Borenstein and Bushnell, 1999). Several different studies, using different data and different empirical techniques have analyzed pricing behavior in California during Summer 2000. They have all come to very similar conclusions. The evidence that there was a significant market power effect reflected in wholesale market

prices in California during Summer 2000 is overwhelming. Indeed, no comprehensive studies exist that come to a different conclusion.³⁴

The measurement of market power is logically separable from the questions of whether and what policymakers should do about it when it is found. The problem that we have focused on here and elsewhere is to develop and apply techniques to measure the presences *and* the magnitude of market power and to understand better the conditions where it is most likely to arise. We recognize that many markets are imperfectly competitive and that it would be fruitless, and probably counterproductive, for policymakers to try to achieve perfectly competitive markets. However, the measurement techniques and applications presented here and elsewhere can be of value to policymakers to determine whether market power problems are sufficiently severe to require some policy response, and if they are, provide some modest guidance to choose among potential structural and behavioral mitigation measures.

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APPENDICES

These Appendices present data underlying our analysis and/or illustrating our methods.

A. Net Import Adjustment

Table A1 below shows actual PX prices, estimated marginal costs using our methods (\$35 RTC price in this case), actual net imports, and estimated competitive net imports for the top 15 load percentiles in August 2000. Our estimates of competitive net imports are below observed net imports when our estimates of marginal costs are below observed PX prices. At the 97th percentile, net imports above those observed are required to clear the market. We use our elasticity relationship to find the price of imports required to induce the capacity needed to clear the market. That price is above the observed PX price and represents the kind of Out of Market (OOM) transaction that the ISO entered into under such conditions.

Table A1. Sample Net Import Calculation for August 2000

Load Percentile	PX Price [1]	MC [2]	Actual NI [3]	Estimated NI [4]
85	232.13	99.33	3675	2769
86	258.92	93.89	4427	3157
87	326.22	102.54	4087	2779
88	234.91	102.14	4345	3292
89	307.11	102.54	4301	2984
90	266.63	112.70	3889	2919
91	314.34	127.53	3682	2726
92	342.45	117.34	4823	3375
93	337.72	152.93	4795	3682
94	356.54	306.80	4786	4552
95	374.26	315.50	4165	3935
96	321.31	306.80	4518	4449
97	373.34	463.72	4008	4308
98	392.13	682.44	3929	4725
99	392.55	1162.45	3648	5238

B. RTC NO_x Credit Prices

A unique characteristic of the RTC program is that while allowances periodically expire, the settlement procedures in the program give NO_x emitters up to two months following the close of the cycle period to reconcile RTC allowances with actual emissions. There is an active market in expired allowances during those two months. It is improper, however, to correlate current prices for electric power with price movements in expired allowances.

Figure C1. NO_x RTC Transactions Expiring on June 30, 2000

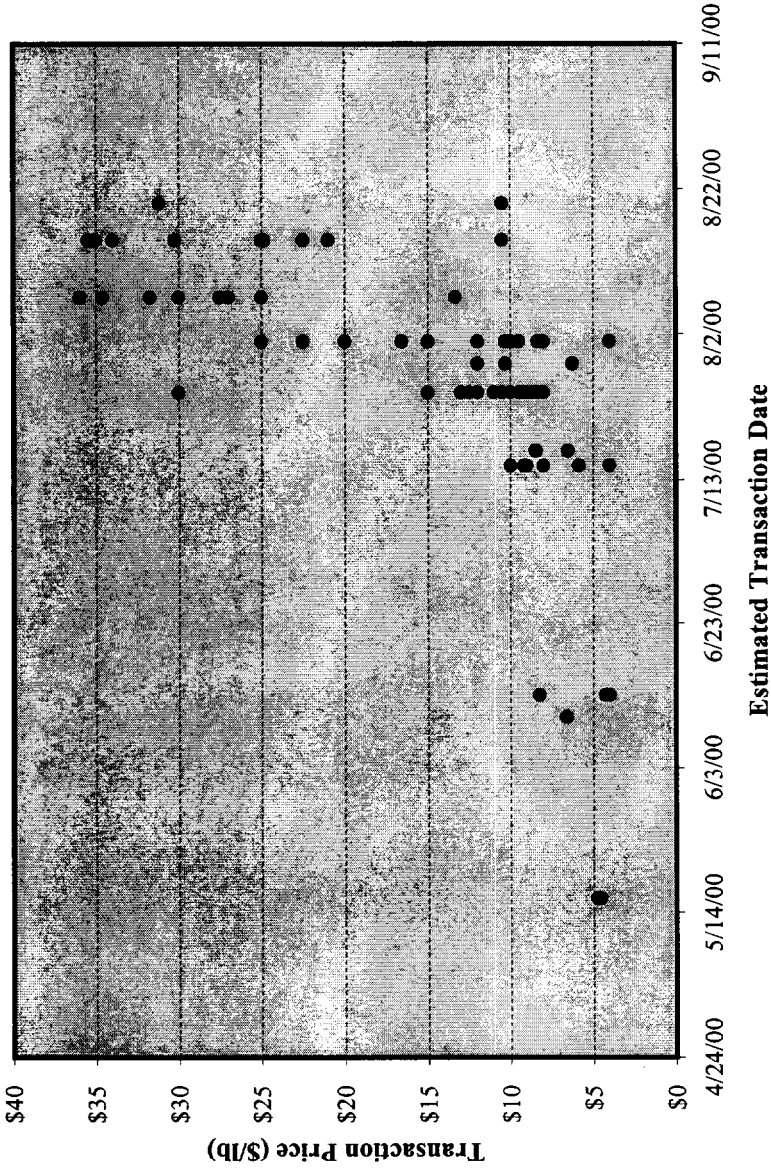
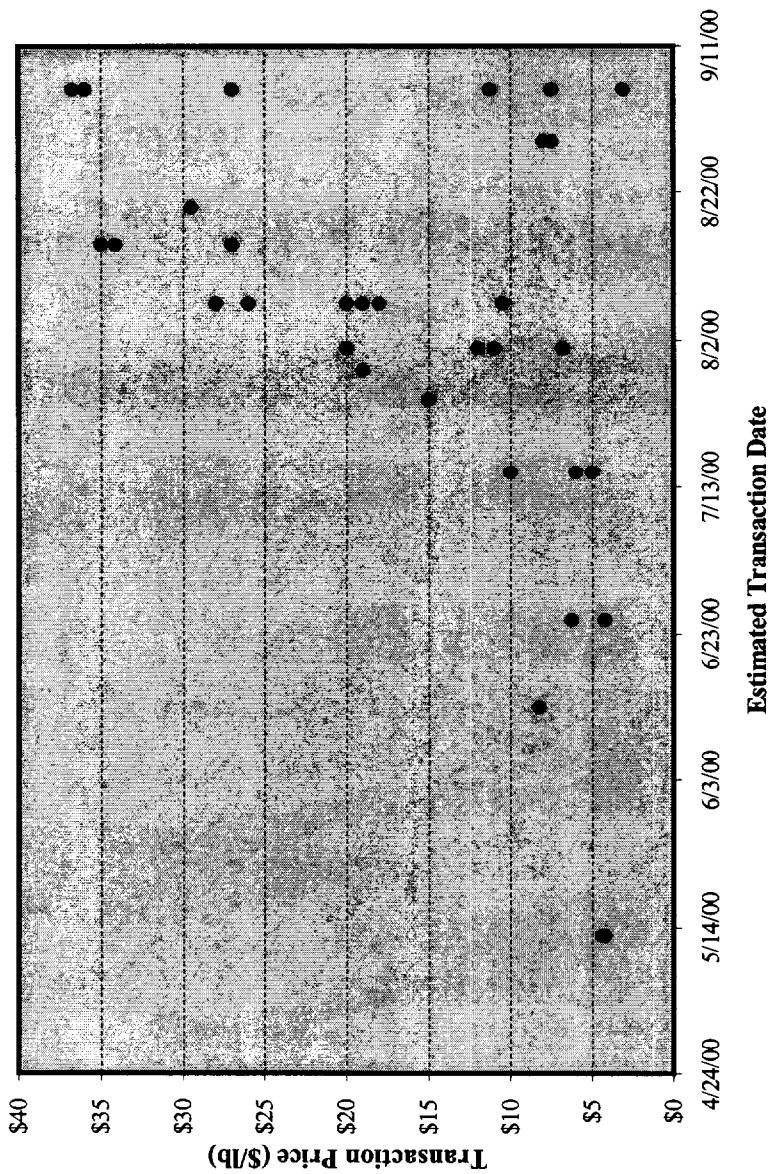


Figure C2. NO_x RTC Transactions Expiring on December 31, 2000



Competitive prices will reflect the marginal costs of current inputs to current generation; competitive prices do not recoup unanticipated increases in sunk costs from past periods. We have therefore examined the prices of RTC NO_x credits over the study period in two groups: prices for June are represented by June prices for RTCs expiring on June 30, 2000, and prices for the post-June period are represented by contemporaneous prices for RTCs expiring on December 31, 2000.

The main data issue for the SCAQMD's list of transactions at more than \$4.00 is that the date given for an observation is the "registration recording date" (RRD) not the date the transaction was executed or received by SCAQMD. We believe that the lag between the RRD and the "deal date" is about 1.5 weeks. It looks like almost all of the RRDs are either Tuesdays or Fridays plus there is a memo in the materials discussing the receipt of a transaction at \$30 on the 27th of July. The RRD for this transaction is the 4th of August.

With that caveat, we reviewed transactions over time for RTCs both for the period ending 6/30/00 and for that ending 12/31/00. The graphs of these transactions by estimated date are shown below. On the basis of these data, we choose \$10/lb as the June RTC price, \$20/lb as the July price and \$35/lb as the August and September prices.